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James Doss-Gollin

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APPOINTMENTS

Rice University	
Assistant Professor, Department of Civil & Environmental Engineering	2021—
Affiliated Faculty, Severe Storm Prediction, Education, and Evacuation from Disasters (SSPEED) Center	2021—
Affiliated Faculty, Ken Kennedy Institute	2023—
The Pennsylvania State University	
Postdoctoral Scholar, Earth & Environmental Systems Institute	2020

EDUCATION

Columbia University	
Ph.D. in Earth & Environmental Engineering	2020
M.S. in Earth & Environmental Engineering	2016
Yale University	
B.S. in Mechanical Engineering	2015

AWARDS AND HONORS

NSF CAREER Award , National Science Foundation	2026
Outstanding Reviewer Award , Earth's Future	2023
Nickolas and Liliana Themelis Fellowship , Fu Foundation School of Engineering and Applied Science, Columbia University	2018
Graduate Research Fellowship, Climate and Large-Scale Atmospheric Dynamics , National Science Foundation	2017
Presidential Distinguished Fellowship , Fu Foundation School of Engineering and Applied Science, Columbia University	2015
Distinction in Major , Department of Mechanical Engineering and Materials Science, Yale University	2015
Legacy Award , New Haven Promise	2015
Larry Coben '79 Fellowship , Yale University	2014
Vance-Carter Travel Award , Yale University	2013
Thomas C. Barry Travel Award , Yale University	2012

ADVISEE AWARDS

Dongwook Kim : Karen and John Huff Graduate Fellowship in Civil and Environmental Engineering.	2025
Yuchen Lu : H.W. Reeves Endowed Scholarship.	2022

PUBLICATIONS

[†] denotes Rice advisee publication. Google Scholar citations: 841, h-index: 14, i10-index: 17.

IN PREP. / UNDER REVIEW

[1] Pollack, A., Benedict, J., Deb, M., Doss-Gollin, J. , Judi, D., Lehman, W., Lutz, N., Reesman, C., Sarazen, E., Son, Y., Srikrishnan, V., Sun, N., and Keller, K. : <i>Unrefined National Building Inventories Can Mislead Risk Assessments and Decisions.</i> . DOI: 10.2139/ssrn.5575271	—
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JOURNAL ARTICLES

- [28] Baer, J. A., Sebastian, A., Grimley, L. E., **Doss-Gollin, J.**, Wright, D. B., Hussain, M. A., and Webber, M. K. : “Neglecting Spatiotemporal Rainfall Variability Misrepresents Flood Hazard and Risk”. *npj Natural Hazards*. DOI: [10.1038/s44304-026-00208-5](https://doi.org/10.1038/s44304-026-00208-5) 2026
- [27] Hancock, C. L., Dee, S. G., Haider, M. R., **Doss-Gollin, J.**, Lehner, F., Murphy, K., and Muñoz, S. E. : “Twenty-First-Century Hydrological Trends in the Mississippi River Basin Intensify the East–West Moisture Gradient”. *Journal of Climate*. DOI: [10.1175/jcli-d-25-0340.1](https://doi.org/10.1175/jcli-d-25-0340.1) 2026
- [26] Murphy, K., Dee, S., Hancock, C., Pitchon, E., **Doss-Gollin, J.**, Wallace, E., and Muñoz, S. : “The Changing Relationship between the Bermuda High, Great Plains Low-Level Jet, and Mississippi River Basin Hydroclimate from the Last Millennium to 2100”. *Journal of Geophysical Research: Atmospheres*. DOI: [10.1029/2025JD044340](https://doi.org/10.1029/2025JD044340) 2026
- [25] O'Donnell, M., Dee, S., **Doss-Gollin, J.**, and Muñoz, S. E. : “Hydrologic Whiplash in the Mississippi River Basin: Mechanisms and Projections”. *Geophysical Research Letters*. DOI: [10.1029/2026GL122807](https://doi.org/10.1029/2026GL122807) 2026
- [24] Pollack, A. B., Auermuller, L., Burleyson, C. D., Campbell, J., Condon, M., Cooper, C., Coronese, M., Dangendorf, S., **Doss-Gollin, J.**, Hegde, P., Helgeson, C., Kopp, R. E., Kwakkel, J., Lesk, C., Mankin, J., Nicholas, R. E., Rice, J., Roth, S., Srikrishnan, V., Scheeler, M., Tuana, N., Vernon, C., Zhao, M., and Keller, K. : “Unlocking the Benefits of Transparent and Reusable Science for Climate Risk Management”. *Proceedings of the National Academy of Sciences*. DOI: [10.1073/pnas.2422157123](https://doi.org/10.1073/pnas.2422157123) 2026
- [23] Haider, M. R., Dee, S., **Doss-Gollin, J.**, Dunne, K., and Muñoz, S. : “Impact of 21st Century Climate Change on Mississippi River Basin Discharge in CESM2 Large Ensemble Projections”. *Global and Planetary Change*. DOI: [10.1016/j.gloplacha.2025.104742](https://doi.org/10.1016/j.gloplacha.2025.104742) 2025
- [22] Liu, C., Kowal, D. R., **Doss-Gollin, J.**, and Vannucci, M. : “Bayesian Functional Graphical Models with Change-Point Detection”. *Computational Statistics and Data Analysis*. DOI: [10.1016/j.csda.2024.108122](https://doi.org/10.1016/j.csda.2024.108122) 2025
- [21] Liu, Y., **Doss-Gollin, J.**, Dai, Q., Veeraraghavan, A., and Balakrishnan, G. : “Downscaling Extreme Precipitation with Wasserstein Regularized Diffusion”. *IEEE Transactions on Geoscience and Remote Sensing*. DOI: [10.1109/TGRS.2025.3611872](https://doi.org/10.1109/TGRS.2025.3611872) 2025
- [20] **Lu, Y.[†]**, Seiyon Lee, B., and **Doss-Gollin, J.**: “Bayesian Spatiotemporal Nonstationary Model Quantifies Robust Increases in Daily Extreme Rainfall across the Western Gulf Coast”. *Environmental Research: Climate*. DOI: [10.1088/2752-5295/adf56e](https://doi.org/10.1088/2752-5295/adf56e) 2025
- [19] O'Donnell, M., Murphy, K., **Doss-Gollin, J.**, Dee, S., and Munoz, S. : “Evaluation of Hydroclimatic Biases in the Community Earth System Model (CESM1) within the Mississippi River Basin”. *Hydrology and Earth System Sciences*. DOI: [10.5194/hess-29-4637-2025](https://doi.org/10.5194/hess-29-4637-2025) 2025
- [18] Pollack, A., **Doss-Gollin, J.**, Srikrishnan, V., and Keller, K. : “UNSAFE: An UNcertain Structure and Fragility Ensemble Framework for Property-Level Flood Risk Estimation”. *Journal of Open Source Software*. DOI: [10.21105/joss.07527](https://doi.org/10.21105/joss.07527) 2025
- [17] Kazadi, A., **Doss-Gollin, J.**, Sebastian, A., and Silva, A. : “FloodGNN-GRU: A Spatio-Temporal Graph Neural Network for Flood Prediction”. *Environmental Data Science*. DOI: [10.1017/eds.2024.19](https://doi.org/10.1017/eds.2024.19) 2024
- [16] Murphy, K., Dee, S., **Doss-Gollin, J.**, Dunne, K. B. J., O'Donnell, M., and Muñoz, S. : “Competing Influences of Land Use and Greenhouse Gas Emissions on Mississippi River Basin Hydroclimate Simulated over the Last Millennium”. *Paleoceanography and Paleoclimatology*. DOI: [10.1029/2024PA004902](https://doi.org/10.1029/2024PA004902) 2024
- [15] Singh, D., Bekris, Y. S., Rogers, C. D. W., **Doss-Gollin, J.**, Coffel, E. D., and Kalashnikov, D. A. : “Enhanced Solar and Wind Potential during Widespread Temperature Extremes across the U.S. Interconnected Energy Grids”. *Environmental Research Letters*. DOI: [10.1088/1748-9326/ad2e72](https://doi.org/10.1088/1748-9326/ad2e72) 2024
- [14] Amonkar, Y., **Doss-Gollin, J.**, Farnham, D. J., Modi, V., and Lall, U. : “Differential Effects of Climate Change on Average and Peak Demand for Heating and Cooling across the Contiguous USA”. *Communications Earth & Environment*. DOI: [10.1038/s43247-023-01048-1](https://doi.org/10.1038/s43247-023-01048-1) 2023
- [13] Amonkar, Y., **Doss-Gollin, J.**, and Lall, U. : “Compound Climate Risk: Diagnosing Clustered Regional Flooding at Inter-Annual and Longer Time Scales”. *Hydrology*. DOI: [10.3390/hydrology10030067](https://doi.org/10.3390/hydrology10030067) 2023
- [12] **Doss-Gollin, J.**, Amonkar, Y., **Schmeltzer, K.[†]**, and Cohan, D. : “Improving the Representation of Climate Risks in Long-Term Electricity Systems Planning: A Critical Review”. *Current Sustainable/Renewable Energy Reports*. DOI: [10.1007/s40518-023-00224-3](https://doi.org/10.1007/s40518-023-00224-3) 2023

- [11] **Doss-Gollin, J.** and Keller, K. : “A Subjective Bayesian Framework for Synthesizing Deep Uncertainties in Climate Risk Management”. *Earth’s Future*. DOI: [10.1029/2022EF003044](https://doi.org/10.1029/2022EF003044) 2023
- [10] Garcia, M., Juan, A., **Doss-Gollin, J.**, and Bedient, P. : “Leveraging Mesh Modularization to Lower the Computational Cost of Localized Updates to Regional 2D Hydrodynamic Model Outputs”. *Engineering Applications of Computational Fluid Mechanics*. DOI: [10.1080/19942060.2023.2225584](https://doi.org/10.1080/19942060.2023.2225584) 2023
- [9] Wutich, A., Thomson, P., Jepson, W., Stoler, J., Cooperman, A. D., **Doss-Gollin, J.**, Jantrania, A., Mayer, A., Nelson-Nuñez, J., Walker, W. S., and Westerhoff, P. : “MAD Water: Integrating Modular, Adaptive, and Decentralized Approaches for Water Security in the Climate Change Era”. *WIREs Water*. DOI: [10.1002/wat2.1680](https://doi.org/10.1002/wat2.1680) 2023
- [8] Zhou, X., Duenas-Osorio, L., **Doss-Gollin, J.**, Liu, L., Stadler, L., and Li, Q. : “Mesoscale Modeling of Distributed Water Systems Enables Policy Search”. *Water Resources Research*. DOI: [10.1029/2022WR033758](https://doi.org/10.1029/2022WR033758) 2023
- [7] **Doss-Gollin, J.**, Farnham, D. J., Lall, U., and Modi, V. : “How Unprecedented Was the February 2021 Texas Cold Snap?”. *Environmental Research Letters*. DOI: [10.1088/1748-9326/ac0278](https://doi.org/10.1088/1748-9326/ac0278) 2021
- [6] **Doss-Gollin, J.**, Farnham, D. J., Ho, M., and Lall, U. : “Adaptation over Fatalism: Leveraging High-Impact Climate Disasters to Boost Societal Resilience”. *Journal of Water Resources Planning and Management*. DOI: [10.1061/\(asce\)wr.1943-5452.0001190](https://doi.org/10.1061/(asce)wr.1943-5452.0001190) 2020
- [5] **Doss-Gollin, J.**, Farnham, D. J., Steinschneider, S., and Lall, U. : “Robust Adaptation to Multiscale Climate Variability”. *Earth’s Future*. DOI: [10.1029/2019ef001154](https://doi.org/10.1029/2019ef001154) 2019
- [4] Rözer, V., Kreibich, H., Schröter, K., Müller, M., Sairam, N., **Doss-Gollin, J.**, Lall, U., and Merz, B. : “Probabilistic Models Significantly Reduce Uncertainty in Hurricane Harvey Pluvial Flood Loss Estimates”. *Earth’s Future*. DOI: [10.1029/2018ef001074](https://doi.org/10.1029/2018ef001074) 2019
- [3] **Doss-Gollin, J.**, Muñoz, Á. G., Mason, S. J., and Pastén, M. : “Heavy Rainfall in Paraguay during the 2015/16 Austral Summer: Causes and Subseasonal-to-Seasonal Predictive Skill”. *Journal of Climate*. DOI: [10.1175/JCLI-D-17-0805.1](https://doi.org/10.1175/JCLI-D-17-0805.1) 2018
- [2] Farnham, D. J., **Doss-Gollin, J.**, and Lall, U. : “Regional Extreme Precipitation Events: Robust Inference from Credibly Simulated GCM Variables”. *Water Resources Research*. DOI: [10.1002/2017WR021318](https://doi.org/10.1002/2017WR021318) 2018
- [1] **Doss-Gollin, J.**, de Souza Filho, F. d. A., and da Silva, F. O. E. : “Analytic Modeling of Rainwater Harvesting in the Brazilian Semi-arid Northeast”. *Journal of the American Water Resources Association*. DOI: [10.1111/1752-1688.12376](https://doi.org/10.1111/1752-1688.12376) 2016

PEER-REVIEWED CONFERENCE PAPERS

- [4] Kazadi, A., **Doss-Gollin, J.**, and Silva, A. : “Pluvial Flood Emulation with Hydraulics-Informed Message Passing”. *Forty-First International Conference on Machine Learning*. URL: <https://openreview.net/forum?id=kIHIA6LrOB> 2024
- [3] Kazadi, A. N., **Doss-Gollin, J.**, Sebastian, A., and Silva, A. : “Flood Prediction with Graph Neural Networks”. *Climate Change AI*. URL: <https://www.climatechange.ai/papers/neurips2022/75> 2022
- [2] Araújo Júnior, L. M., de Souza Filho, F. d. A., da Silva Silveira, C., Aragão Dias, T., and **Doss-Gollin, J.** : “Análise Dos Eventos de Seca No Nordeste Setentrional Brasileiro Com Base No Índice de Precipitação Normalizada”. *XII Simpósio de Recursos Hídricos Do Nordeste*. DOI: [10.13140/rg.2.1.4610.7685](https://doi.org/10.13140/rg.2.1.4610.7685) 2014
- [1] **Doss-Gollin, J.**, de Souza Filho, F. d. A., and da Silva, F. O. E. : “Considerações Sobre a Sustentabilidade Hídrica de Cisternas Para Captação de Chuva No Semiárido Brasileiro”. *XII Simpósio de Recursos Hídricos Do Nordeste*. DOI: [10.13140/rg.2.1.4086.4807](https://doi.org/10.13140/rg.2.1.4086.4807) 2014

SPONSORED RESEARCH

Amounts reflect Rice portion for collaborative grants and subawards; total amount for direct awards.

- PI, National Science Foundation – Infrastructure Systems and People. *CAREER: Drain or Retain? Strategic Assessment of Integrated Flood Infrastructure Portfolios with Physics-Informed ML*. No. 2543303. (\$599,581) 2026–2031

Co-PI, National Science Foundation – Confronting Hazards, Impacts and Risks for a Resilient Planet (CHIRRP). <i>RAISE: Flood Resilience in Rural Texas Communities</i> PI: Avantika Gori, with Andrew Juan and Keri Stephens. No. 2440167. (\$999,986)	2025–2028
PI, Consortium for Enhancing Resilience and Catastrophe Modeling (CERCAT). <i>A Nonstationary Joint Probability Method for Tropical Cyclone Hazard Assessment</i> with Avantika Gori. No. P12. (\$75,000)	2025–2026
PI, Ken Kennedy Institute at Rice University. <i>Advancing AI for Climate Risk and Urban Resilience</i> with Sylvia Dee, Avantika Gori, Arlei Lopes da Silva, Jamie Padgett, and Noemi Vergopolan. (\$80,000)	2025–2026
Co-PI, NVIDIA. <i>Computing Infrastructure for AI-enhanced Climate Risk and Resilience at Rice</i> PI: Arlei Lopes da Silva, with Avantika Gori. in-kind	2025–2025
PI, Ken Kennedy Institute at Rice University. <i>Advancing AI for Climate Risk and Urban Resilience</i> with Sylvia Dee, Avantika Gori, Arlei Lopes da Silva, Jamie Padgett, and Noemi Vergopolan. (\$80,000)	2024–2025
Co-PI, National Science Foundation – Industry-University Cooperative Research Centers (IUCRC). <i>IUCRC Planning Grant Rice University: Center for Climate, Equity and Resilience in Catmodeling (CER-Cat)</i> PI: Jamie Padgett, with Katherine Ensor, Avantika Gori, Leonardo Dueñas Osorio, Brian Davison, David Casagrande, Maryam Rahnemoonfar, Paolo Bocchini, and Yi-Chen Yang. No. 2413382. (\$20,000)	2024–2025
Co-PI, Rice University Sustainability Institute. <i>Workshop on Nature-Based Solutions for Resilient Coastal Cities</i> PI: Philip Bedient, with Avantika Gori. (\$7,500)	2024–2025
PI, Texas Water Development Board. <i>Developing Future Rainfall Frequency Grids for the State of Texas</i> with John Nielsen-Gammon. (Full grant \$192,828. Subaward \$77,750)	2022–2026
Co-PI, National Science Foundation – Climate and Large-Scale Dynamics (CLD). <i>Collaborative Research: Evaluating the Past and Future of Mississippi River Hydroclimatology to Constrain Risk via Integrated Climate Modeling, Observations, and Reconstructions</i> PI: Sylvia Dee, with Samuel Muñoz. No. 2147781. (\$472,024)	2022–2025
PI, National Science Foundation – Strengthening America’s Infrastructure (SAI). <i>Collaborative Research: EAGER: Participatory Design for Water Quality Monitoring of Highly Decentralized Water Infrastructure Systems</i> with Alicia Cooperman, Alex Mayer, Shane Walker, and Manuela Muñoz Fuerte. No. 2120829. (\$85,046)	2022–2025
Co-PI, 100,000 Strong in the Americas Innovation Fund. <i>IFCE-Rice-SENAI Program on Artificial Intelligence for Urban Sustainability and Resilience to Natural Disasters in the Americas</i> PI: Arlei Lopes da Silva. (\$50,000)	2022–2023
Co-PI, Energy Foundation. <i>Synthesis of Texas Electricity Research from Rice University</i> PI: Daniel Cohan. (\$24,928)	2022–2023
PI, Rice University – Sustainable Futures Fund (SFF). <i>Leveraging Earth System Observations at Multiple Scales to Improve Stormwater Management in Houston</i> with Sylvia Dee. (\$50,000)	2022–2023

PRESENTATIONS

INVITED TALKS

“Statistical Modeling for Nonstationary Hydroclimate Hazards: Challenges and Opportunities”. Statistics Colloquium, Rice University , Houston, TX.	2025
Panelist. Exploring Water Resilience to Emerging Challenges, Society for Risk Analysis , Remote Presentation.	2025
“Advancing Urban Flood Risk Management through Physics-informed, Data-Driven Hazard Assessment”. Earth, Marine, and Environmental Science Seminar, University of North Carolina , Chapel Hill, NC.	2024
“Quantifying and Characterizing Uncertain Climate Hazards to Enable Adaptive Resilience”. Atmospheric Sciences Seminar, Texas A&M University , College Station, TX.	2022
“Revisiting Our Design Criteria: What Hazards Should We Design for in a Changing Climate?”. Hydrologic Sciences and Water Resources Engineering Seminar, University of Colorado Boulder , Remote Presentation.	2022
“Adapting Engineering Design Criteria to a Changing Climate: Insights from House Elevation”. Technical Webinar, ASCE Central New Jersey Branch , Remote Presentation.	2022
Panelist. Extreme Weather: How To Report on a World That’s Warmer, Colder, Wetter, Drier and Weirder, 31st Annual Conference of the Society of Environmental Journalists , Houston, TX.	2022

"Predictable and Preventable: Lessons in Climate Risk Management from the Texas Coast (and Beyond)". Climate Risk and Financial Institutions Seminar (Prof. Madison Condon), Boston University School of Law , Remote Presentation.	2022
"Extreme Impacts Don't Require Extreme Weather: Lessons from the February 2021 Texas Blackouts". Outreach Event: Science is for Everyone, American Meteorological Society , Remote Presentation.	2022
"Extreme Impacts Don't Require Extreme Weather: Lessons from the February 2021 Texas Blackouts". Compound Events Working Group, Risk KAN (Knowledge Action Networks) , Remote Presentation.	2021
Panelist. Tail events: Prediction, Planning, and Performance, Harvard Electricity Policy Group , Remote Presentation.	2021
"Towards Adaptive Resilience: Managing Flood Risks in a Changing World". Technical Webinar, ASCE Central New Jersey Branch , Remote Presentation.	2021
"Prediction and Implications of Structured Climate Risk for Sequential Adaptation under Deep Uncertainty". Center for Climate Risk Management CLIMA Seminar, the Pennsylvania State University , State College, PA.	2020
"Prediction and Implications of Structured Climate Risk for Sequential Adaptation under Deep Uncertainty". Department of Civil and Environmental Engineering Seminar, Rice University , Houston, TX.	2020
"Prediction and Implications of Structured Climate Risk for Sequential Adaptation under Deep Uncertainty". Complex Systems Simulation and Optimization Group, National Renewable Energy Laboratory , Golden, CO.	2020
"Drivers of Extreme Rainfall: Atmospheric Circulation Patterns and Regional Intense Rainfall in the Ohio River". European Flood Awareness System Group, European Centre for Medium Range Weather Forecasting , Reading, England.	2016
"Understanding the Physical Drivers of Extreme Rainfall for Flood Prediction". Oxford Water Network, Oxford University , Oxford, England.	2016

FIRST-AUTHOR CONFERENCE AND WORKSHOP PRESENTATIONS

"Industry-Academia Collaborations on Seasonal and Long-Term TC Hazard Assessment (CERCat)". Catastrophe Risk Management Conference (CATRISK26) , <i>Reinsurance Association of America</i> , Orlando, FL [Panel].	2026
"H24A-07: TxRain: A Bayesian framework for integrating historical observations and model projections to develop nonstationary IDF curves". American Geophysical Union Fall Meeting 2025 , New Orleans, LA [Talk].	2025
"Robust Trends in Extreme Rainfall Probabilities in Texas". 2025 Texas Climate Conference , <i>Rice University and Texas A&M University</i> , Houston, TX [Talk].	2025
"Use-Inspired Tools for Climate Hazard Assessment". Nature-Based Solutions for a Resilient Gulf Coast Workshop , <i>Rice University</i> , Houston, TX [Talk].	2025
"Advancing Urban Flood Hazard Characterization through Machine Learning: Challenges and Opportunities". American Geophysical Union Fall Meeting 2024 , Washington, DC [Talk].	2024
"Assessing and Managing Climate Risks to Electricity Systems in an Era of Climate Change and Energy Transition". American Geophysical Union Fall Meeting 2024 , Washington, DC [Talk].	2024
"Leveraging Machine Learning to Advance Urban Flood Hazard Assessment: Challenges and Opportunities". Data-driven and physics-based machine learning methods for forecasting and knowledge discovery of surface hydrology , <i>Conference on Computational Methods in Water Resources</i> , Tucson, AZ [Talk].	2024
"Incorporating Temperature Projections into Energy Systems Planning". Extreme Heat Workshop , <i>Columbia University</i> , New York, NY [Talk].	2024
"NH14B-07: Linking Robust Trends in Observations and Models to Develop Nonstationary Rainfall Frequency Grids for the State of Texas". American Geophysical Union Fall Meeting 2023 , San Francisco, CA [Talk].	2023
"A Bayesian Spatial Hierarchical Framework for Process-Informed Nonstationary Analysis of Precipitation Frequencies". 13th International Workshop on Statistical Hydrology , <i>International Association of Hydrological Sciences</i> , Boston, MA [Talk].	2023

"H35F-07: Near-term Predictability Lowers Long-Term Adaptation Costs". American Geophysical Union Fall Meeting 2022 , Chicago, IL [Talk].	2022
"Unprecedented Impacts Don't Require Unprecedented Weather". Post-Harvey Climate & Flood Impacts on the Built Environment , <i>Severe Storm Prediction, Education, & Evacuation from Disasters Center</i> , Houston, TX.	2022
"H25U-1265: Operationalizing Bayesian Model Checking for Robust Decision Making: Insights from House Elevation". American Geophysical Union Fall Meeting 2021 , New Orleans, LA [Poster].	2021
"A14H-03: How Unprecedented Was the February 2021 Texas Cold Snap?". American Geophysical Union Fall Meeting 2021 , New Orleans, LA [Talk].	2021
"Valuing Flexibility and Soft Instruments for Sequential Decision Problems". 2020 Annual Meeting, Society for Decision Making under Deep Uncertainty , [Remote Presentation].	2020
"H11G-07: Towards Adaptive Resilience: Managing Uncertainties and Exploiting Predictability across Timescales". American Geophysical Union Fall Meeting 2019 , San Francisco, CA [Talk].	2019
"Adaptive Resilience through Real Options and Deep Reinforcement Learning". Doctoral Consortium on Computational Sustainability , <i>Carnegie Mellon University</i> , Pittsburgh, PA [Talk].	2019
"Evaluating Staged Investments in Critical Infrastructure for Climate Adaptation". 2019 Interdisciplinary Ph.D. Workshop in Sustainable Development , <i>Columbia University</i> , New York, NY [Talk].	2019
"H52F-05: Robust Adaptation to Cyclical Climate Risk". American Geophysical Union Fall Meeting 2018 , Washington, DC [Talk].	2018
"Robust Adaptation to Multi-Scale Climate Variability". The Nexus of Climate Data, Insurance, and Adaptive Capacity , Asheville, NC [Poster].	2018
"A31H-0135: Causes and Model Skill of the Persistent Intense Rainfall and Flooding in Paraguay during the Austral Summer 2015-2016". American Geophysical Union Fall Meeting 2017 , New Orleans, LA [Talk].	2017
"H22B-02: Designing and Operating Infrastructure for Nonstationary Flood Risk Management". American Geophysical Union Fall Meeting 2017 , New Orleans, LA [Talk].	2017
"Extreme Rainfall in Paraguay during the 2015-16 Austral Summer: Causes and Predictive Skill". North East Graduate Student Water Symposium , <i>University of Massachusetts Amherst</i> , Amherst, MA [Talk].	2017
"Regional Intense Precipitation: Inferences From GCM Atmospheric Circulation Fields". Modeling Research in the Cloud , <i>National Center for Atmospheric Research</i> , Boulder, CO [Poster].	2017
"Statistical-Dynamical Analysis of Climate Projections for Flood Infrastructure Design". Interdisciplinary Ph.D. Workshop in Sustainable Development 2017 , <i>Columbia University</i> , New York, NY [Talk].	2017
"Causes and Model Skill of the Persistent Intense Rainfall and Flooding in Paraguay during the Austral Summer 2015-2016". Workshop on Subseasonal to Seasonal Predictability of Extreme Weather and Climate , <i>Columbia University</i> , New York, NY [Poster].	2016

ADVISEE CONFERENCE AND WORKSHOP PRESENTATIONS

Dongwook Kim [†] : "Evaluating the Generalization of Sentinel-1 Flood Mapping across Diverse Environments". Fall Meeting , <i>Consortium for Enhancing Resilience and Catastrophe Modeling (CERCat)</i> , Bethlehem, PA [Poster].	2026
Dongwook Kim [†] : "Comparative Analysis of Satellite-Based Flood Mapping: A Case Study of Hurricane Harvey". 39th Annual International Conference on Coastal Engineering (ICCE) , <i>Coasts, Oceans, Ports & Rivers Institute (COPRI) - ASCE</i> , Galveston, TX [Talk].	2026
Yuchen Lu [†] : "NH13D-0410: A Hierarchical Bayesian Spatial Framework to Assess Nonstationary Rainfall Intensity, Frequency, and Duration in Texas". American Geophysical Union Fall Meeting 2025 , New Orleans, LA [Poster].	2025
Yuchen Lu [†] : "TxRAIN-Observational: A Hierarchical Bayesian Spatial Framework to Assess Nonstationary Rainfall Intensity, Frequency, and Duration in Texas". American Geophysical Union Fall Meeting 2024 , Washington, DC [Talk].	2024
Yuchen Lu [†] : "Nonstationary Extreme Precipitation Probabilities in Texas". Fall Meeting , <i>Consortium for Enhancing Resilience and Catastrophe Modeling (CERCat)</i> , Bethlehem, PA [Poster].	2024

Yuchen Lu[†] : “Spatially Varying and Duration Dependent Covariate Model: A Hierarchical Bayesian Framework for Multi-duration Extreme Precipitation Frequency Analysis in Texas”. World Environmental & Water Resources Congress 2024 , Environmental & Water Resources Institute (EWRI) - ASCE, Milwaukee, WI [Talk].	2024
Yuchen Lu[†] : “H21T-1602: Spatially Varying Covariate Model: A Hierarchical Bayesian Framework for Precipitation Frequency Analysis in the Gulf Coast”. American Geophysical Union Fall Meeting 2023 , San Francisco, CA [Talk].	2023
Yuchen Lu[†] : “H42E-1333: Nonstationary GEV with Hierarchical Spatial Pooling: A Spatiotemporal Bayesian Framework for Nonstationary Extreme Precipitation Frequency Analysis in the Gulf Coast”. American Geophysical Union Fall Meeting 2022 , Chicago, IL [Talk].	2022

TEACHING AND ADVISING

COURSES TAUGHT

Rice CEVE 543: <i>Statistical-Physical Methods for Hydroclimate Extremes and Catastrophes</i> .	Fall 2026
Rice CEVE 421/521: <i>Climate Risk Management</i> .	Spring 2026
Rice CEVE 543: <i>Statistical-Physical Methods for Hydroclimate Extremes and Catastrophes</i> .	Fall 2025
Rice CEVE 101: <i>Fundamentals of Civil and Environmental Engineering</i> .	Fall 2024
Rice CEVE 421/521: <i>Climate Risk Management</i> .	Spring 2024
Rice CEVE 543: <i>Data Science Methods for Climate Hazard Assessment</i> .	Fall 2023
Rice CEVE 421/521: <i>Climate Risk Management</i> .	Spring 2023
Rice CEVE 543: <i>Environmental Data Science</i> .	Spring 2022
Rice CEVE 101: <i>Fundamentals of Civil and Environmental Engineering</i> .	Fall 2021

CURRENT GRADUATE STUDENTS

Andres Calvo: Ph.D. in Civil and Environmental Engineering, Rice University. Committee Member .	
True Furrh: Ph.D. in Civil and Environmental Engineering, Rice University. Committee Member .	2021–
Dongwook Kim: Ph.D. in Civil and Environmental Engineering, Rice University. Primary Advisor .	2024–
Kelsey Murphy: Ph.D. in Earth, Environmental, and Planetary Sciences, Rice University. Committee Member .	2021–
Jiayue Yin: Ph.D. in Earth, Environmental, and Planetary Sciences, Rice University. Committee Member .	2023–

PAST GRADUATE STUDENTS

Primary Advisor

Yuchen Lu: Ph.D. in Civil and Environmental Engineering, Rice University. Thesis: <i>Probabilistic Analysis of Nonstationary Hydroclimate Extremes</i> .	2026
Katlyn Schmeltzer: MCEE in Civil and Environmental Engineering, Rice University. Thesis: <i>Pre-Trained Long Short-Term Memory Network Performance for Streamflow Prediction in the Brazos River Basin</i> .	2025

Committee Member

Valeriia Sobolevskaia: Ph.D. in Earth, Environmental, and Planetary Sciences, Rice University. Thesis: <i>The seismic signatures of the hydrological cycle as observed by distributed acoustic sensing</i> .	2026
Karan Jakhar: Ph.D. in Mechanical Engineering, Rice University. Thesis: <i>Equation Discovery and Deep Learning for Geophysical Turbulence</i> .	2025
Kyle Ostlind: M.S. in Civil and Environmental Engineering, Rice University. Thesis: <i>Evaluating Runoff Response to Nature-Based Solutions under Varying Development Scenarios in Upper Cypress Creek near Houston, Texas</i> .	2025
John A. Baer: M.S. in Earth, Marine and Environmental Sciences, University of North Carolina at Chapel Hill. Thesis: <i>Quantifying Precipitation-Induced Uncertainty in Flood Hazard Assessment in a Coastal Urban Area</i> .	2024
Kendall Capshaw: Ph.D. in Civil and Environmental Engineering, Rice University. Thesis: <i>Modeling Coastal Petrochemical Infrastructure Risk, Resilience, and Cascading Community Consequences</i> .	2024
Xinyue Luo: Ph.D. in Earth, Environmental, and Planetary Sciences, Rice University. Thesis: <i>Characterizing the El Niño-Southern Oscillation and Its North American Teleconnections over the Last Millennium</i> .	2024

Anibal Tafur Gutierrez: Ph.D. in Civil and Environmental Engineering, Rice University. Thesis: <i>Methods and Tools for Risk-Informed Resilience Enhancement of Coastal Intermodal Freight Networks</i> .	2024
Matthew Garcia: Ph.D. in Civil and Environmental Engineering, Rice University. Thesis: <i>Novel Urban Floodplain Modeling Methods for Applications in Coupling Surrogate Machine Learning Methods</i> .	2023
Mia Peebles: M.S. in Civil and Environmental Engineering, Rice University. Thesis: <i>Modeling Flood Reduction of Nature-Based Channel Modifications in Houston, TX</i> .	2023
Xiangnan Zhou: Ph.D. in Civil and Environmental Engineering, Rice University. Thesis: <i>Resilience Planning for Water Distribution Systems</i> .	2023
Raychel Bahnick: M.S. in Civil and Environmental Engineering, Rice University. Thesis: <i>Assessing Land Use Change and Subsidence Impact on Inland Flooding</i> .	2022
Alyssa Graham: M.S. in Civil and Environmental Engineering, Rice University. Thesis: <i>Water Supply Vulnerability Testing and Robust Planning Analysis with Exploratory Modeling under Deep Uncertainty</i> .	2022
Elizabeth Hoffmann: M.S. in Civil and Environmental Engineering, Rice University. Thesis: <i>Mapping Dynamic Watershed Response Under Increasing Development Using HEC-RAS 2D: A Case Study of the Big Creek Watershed in Fort Bend County</i> .	2022
Chunshan Liu: Ph.D. in Statistics, Rice University. Thesis: <i>Bayesian Graphical Models for Multivariate Time Series</i> .	2022
Xiaoyu (Toby) Li: M.S. in Civil and Environmental Engineering, Rice University. Thesis: <i>Evaluating the Effects of Project Brays Mitigation Using Unsteady HEC-RAS Hydraulic Modeling: Application to Meyerland in Houston, TX</i> .	2021

UNDERGRADUATE RESEARCHERS

Year indicates graduation year.

Zain Rahman: <i>B.S. in Computer Science</i> , Rice University.	2027
Kyle Olcott: <i>B.S. in Civil and Environmental Engineering</i> , Rice University.	2025
Sophia Prieto: <i>B.S. in Statistics</i> , Rice University.	2023
John Cook: <i>B.S. in Civil and Environmental Engineering</i> , Rice University.	2022

TEAMS ADVISED

Optimal Policy for Decentralized Wastewater Systems (DWS) while Relaxing Certainty: Rice Computational Mathematics and Operations Research (CMOR) Senior Project.	2024-2025
Flood Sight Advancing Real-Time Flood Predictions for Situational Awareness: Rice Data to Knowledge (D2K) Lab.	Spring 2025
Predicting Streamflow for Hill Country Gauges Using a Transformer-based Model: Rice Data to Knowledge (D2K) Lab.	Fall 2025

SERVICE ACTIVITIES

DEPARTMENTAL SERVICE

Member , Graduate Studies Committee.	2024–
Member , Diversity, Equity, and Inclusion Committee.	2024
Member , Faculty Search Committee.	2022–2023
Member , Seminar Committee.	2022–2023

UNIVERSITY SERVICE

Member , Research Council. Ken Kennedy Institute.	2024–
Faculty Associate . Duncan College.	2023–
External Search Committee Member . Department of Earth, Environmental, and Planetary Sciences.	2022–2023

PROFESSIONAL SERVICE

Working Group Member , Digital Twins - Beyond Physical Systems. Digital Earths Lighthouse Activity.	2026–2030
Committee Member , Water and Society Newsletter Committee. American Geophysical Union.	2024–
Committee Member , Environmental and Water Resources Systems (EWRS) Committee. American Society of Civil Engineers (ASCE) Environmental & Water Resources Institute (EWRI).	2021–

PEER REVIEW

Journals: AGU Advances; Climate Risk Management; Climatic Change; Communications Earth and Environment; Earth's Future; Energy Technology; Engineering Applications of Computational Fluid Mechanics; Environmental Data Science; Environmental Research Letters; Environmental Research: Climate; Geophysical Research Letters; Hydrology and Earth System Sciences; IEEE Transactions on Geoscience and Remote Sensing; Joule; Journal of Applied Meteorology and Climatology; Journal of Catastrophe Risk and Resilience; Journal of Hydrology; Journal of Water Resources Planning and Management; Machine Learning: Earth; Natural Hazards and Earth System Sciences; npj Natural Hazards; Oxford Development Studies; Water Resources Research; Water Security; Weather, Climate, and Society.

Funding Agencies: Department of Energy (BER); Dutch Research Council (NWO); National Science Foundation.

Other: Electric Power Research Institute (EPRI); Texas Water Development Board (TWDB).

SESSIONS CONVENED

Co-Convenor. NH43C - <i>Advances in Urban Flood Risk Assessment and Adaptation IV Poster. American Geophysical Union Fall Meeting</i> , New Orleans, LA.	2025
Co-Convenor. NH43D - <i>Advances in Urban Flood Risk Assessment and Adaptation V Poster. American Geophysical Union Fall Meeting</i> , New Orleans, LA.	2025
Co-Convenor. NH41E - <i>Interdisciplinary Advances in Catastrophe Modeling and Disaster Resilience: Bridging Science, Policy, and Practice I Poster. American Geophysical Union Fall Meeting</i> , New Orleans, LA.	2025
Co-Convenor. NH43B - <i>Interdisciplinary Advances in Catastrophe Modeling and Disaster Resilience: Bridging Science, Policy, and Practice II Oral. American Geophysical Union Fall Meeting</i> , New Orleans, LA.	2025
Co-Convenor. NH31A - <i>Advances in Urban Flood Risk Assessment and Adaptation I Oral. American Geophysical Union Fall Meeting</i> , New Orleans, LA.	2025
Co-Convenor. NH32A - <i>Advances in Urban Flood Risk Assessment and Adaptation II Oral. American Geophysical Union Fall Meeting</i> , New Orleans, LA.	2025
Co-Convenor. NH33A - <i>Advances in Urban Flood Risk Assessment and Adaptation III Oral. American Geophysical Union Fall Meeting</i> , New Orleans, LA.	2025
Co-Organizer. <i>Nature-Based Solutions for a Resilient Gulf Coast. Rice University</i> , Houston, TX.	2025
Primary Convenor. H31G - <i>Integrating Social, Scientific, and Engineering Approaches to Identify and Address Gaps in Water Infrastructure and Household Water Security I Oral. American Geophysical Union Fall Meeting</i> , San Francisco, CA.	2023
Primary Convenor. H33P - <i>Integrating Social, Scientific, and Engineering Approaches to Identify and Address Gaps in Water Infrastructure and Household Water Security II Poster. American Geophysical Union Fall Meeting</i> , San Francisco, CA.	2023
Convenor. NH41C - <i>Hybrid Modeling and Digital Twin Systems for Flood Hazard Prediction and Risk Assessment at Different Spatial Scales. American Geophysical Union Fall Meeting</i> , Washington, DC.	2023
Chair. H44G - <i>Water and Society: Interdisciplinary Perspectives on Hydroclimatic Forecasting for Water Resources Decision Making I Oral. American Geophysical Union Fall Meeting</i> , New Orleans, LA.	2021
Chair. H45V - <i>Water and Society: Interdisciplinary Perspectives on Hydroclimatic Forecasting for Water Resources Decision Making III Poster. American Geophysical Union Fall Meeting</i> , New Orleans, LA.	2021
Chair. H45V - <i>Water and Society: Interdisciplinary Perspectives on Hydroclimatic Forecasting for Water Resources Decision Making Poster Summary Session. American Geophysical Union Fall Meeting</i> , New Orleans, LA.	2021
Primary Convenor. NH53 - <i>Emerging Needs and Approaches for Climate Services: Understanding and Developing Innovative Approaches to User-Oriented Climate Services. American Geophysical Union Fall Meeting</i> , San Francisco, CA.	2019
Student Organizer. <i>Earth and Environmental Engineering Student Research Symposium. Columbia University</i> , New York, NY.	2018

Student Organizer. *Earth and Environmental Engineering Student Research Symposium.* **Columbia University,** New York, NY. 2017

MEDIA HIGHLIGHTS

"FEMA elabora nuevos mapas de riesgo de inundación ¿Afectará esto tu propiedad?". **Telemundo Houston** [recorded video interview]. 2026

"The Climate-Disaster Scores That Could Make or Break Your Home Sale". **The Wall Street Journal** — Jean Eaglesham and Nicole Friedman [print]. 2026

"Historic Texas Flooding". **KEYE-AUS (CBS)** [recorded video interview]. 2025

"Houston's Morning Show". **Fox26 Houston** [live video interview]. 2025

"Breaking down the force of water in the Texas floods". **AP News** — Michael Phillis [print]. 2025

"Here are some things you can do to be better prepared for major flooding". **Associated Press** — Caleigh Wells [print]. 2025

"ABC13-Luke Jones speaks with James Doss-Gollin, an assistant professor of Civil and Environmental Engineering at Rice University, about the key differences between flooding in Houston and central Texas." **ABC13** — Luke Jones [recorded video interview]. 2025

"After deadly flooding in Central Texas, state lawmakers look to prevent similar tragedies". **KVUE** — Daniel Perreault [recorded video interview]. 2025

"New Flood Warning System Greenlit Shortly Before Deadly Texas Disaster". **The Wall Street Journal** — Joseph De Avila [print]. 2025

"Questions arise on how emergency warning systems work after Central Texas flood". **ABC13** — Tom Abrahams [print]. 2025

"Bajo la Amenaza del Golfo (Under the Threat of the Gulf)". **Telemundo Houston** [recorded video interview]. 2025

"Will Texas become too hot for humans?". **BBC Future** — Sarah Griffiths [print]. 2023

"Climate change has sent temperatures soaring in Texas". **The Texas Tribune** — Erin Douglas, Yuriko Schumacher and Alex Ford [print]. 2023

"Texas could have foreseen 2021 cold-wave disaster, new study concludes". **Texas Climate News** — Bob Henson [print]. 2022

"Opinion: The risks of climate change are great - so are the rewards of solving it". **Houston Chronicle** — Andrew Dessler, James Doss-Gollin, and Katherine Hayhoe [print]. 2021

"The False Comfort of Higher Seawalls". **The New Republic** — Paola Rosa-Aquino [print]. 2019

"New Study Shows Promise for Long-Term Weather Forecasts in South America". **State of the Planet** — Elisabeth Gawthrop [print]. 2018

LANGUAGE SKILLS

English: Native
Spanish: Fluent
Portuguese: Proficient

TRAINING AND EXPERIENCE

Strategic Program to Accelerate Researchers in Computing (SPARC) Participant — *Natural Hazards Engineering Research Infrastructure (NHERI) DesignSafe-CI.* 2025

Panel Fellow — *NSF CMMI's Game Changer Academies for Advancing Research Innovation.* 2021

Visiting Graduate Researcher — *Lamontagne Research Group, Department of Civil and Environmental Engineering, Tufts University, Medford, MA.* 2019–2020

Summer School Participant — *Fluid Dynamics of Sustainability and the Environment, Cambridge University, Cambridge, England.* 2016

Education Policy Intern — *Elm City Communities / New Haven Housing Authority, New Haven, CT.* 2015

President (2014), Design Lead (2013), Member (2012, 2015) — *Engineers Without Borders, Yale Student Chapter, New Haven, CT.* 2012–2015

Undergraduate Research Assistant — *Lab of Jaehong Kim, Department of Chemical and Environmental Engineering, Yale University, New Haven, CT.* 2014–2015

Visiting Undergraduate Researcher — <i>Water and Climate Risk Lab, Department of Hydraulic and Environmental Engineering, Universidade Federal do Ceará, Fortaleza, Brazil.</i>	2014
Mechanical Design Intern — <i>Slingshot Team, DEKA Research \& Development, Manchester, NH.</i>	2013
Undergraduate Research Assistant — <i>Lab of Jan Schroers, Department of Mechanical Engineering and Materials Science, Yale University, New Haven, CT.</i>	2012
Ikatú Agua Intern — <i>Fundación Paraguaya, Asunción, Paraguay.</i>	2012